

AstraNotes — Web UI Design Document

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1. Purpose

This document defines the design for a Flask-based web UI layer added to AstraNotes for the final project demo. The web UI is a new presentation layer only — it wires directly to the existing `NoteService` without modifying any core business logic, validation, or storage code. This is consistent with NFR-02: the architecture separates presentation from business logic so each can be replaced independently.

2. Scope — What the UI Does and Does Not Include

In scope:

FR	Feature	UI Element
FR-01	Create a note with title, body, tags	Create Note form with submit button
FR-02	Edit an existing note	Edit button on each note card → inline form
FR-03	Delete a note	Delete button with confirmation dialog
FR-06	Show metadata: UUID, timestamps, tags, private flag	Note card footer showing all metadata
FR-07	Keyword search across title and body	Search bar at top of note list, live results
FR-08	Notes sorted by last-modified, newest first	Default list order on page load
FR-04	Private flag (no encryption — deferred per traceability matrix)	Private checkbox on note card, private checkbox on create form

Out of scope:

User accounts and authentication — not in requirements (single-user local app). Sharing or collaboration — not in requirements. Fernet encryption of private notes — deferred slice, documented in traceability matrix as Gap 1. Cloud storage or sync — no network dependency per NFR-02.

3. Architecture

The web UI adds one new package: `src/astranotes/web/`. It contains a single Flask application file and a `templates` folder. No existing packages are modified.

```
AstraNotes/  
■■■ src/astranotes/
```

- ■■■ cli/ ← unchanged
- ■■■ service/ ← unchanged (NoteService)
- ■■■ repository/ ← unchanged (JsonFileRepository)
- ■■■ validation/ ← unchanged (ValidationLayer)
- ■■■ models/ ← unchanged (Note, exceptions)
- ■■■ privacy/ ← unchanged (deferred stub)
- ■■■ web/ ← NEW
- ■■■ __init__.py
- ■■■ app.py ← Flask routes, builds NoteService
- ■■■ templates/
- ■■■ index.html ← single-page UI
- ■■■ tests/ ← unchanged

The Flask app instantiates NoteService the same way cli/main.py does — using JsonFileRepository pointed at ~/.astranotes/data/. Both the CLI and the web UI share the same note store on disk.

4. Flask API Routes

Method	Route	Action	Traces To
GET	/	Load all notes sorted by modified_at desc	FR-08
GET	/?q=keyword	Search notes by keyword	FR-07
POST	/notes	Create a new note (title, body, tags, is_private)	FR-01, FR-06
POST	/notes/<id>/edit	Edit title and body of existing note	FR-02
POST	/notes/<id>/delete	Delete note by ID	FR-03

All routes return redirects or JSON. No external API framework needed — plain Flask with HTML form posts. This keeps the dependency surface minimal (SPR-04).

5. UI Layout

Single-page design — one HTML template (index.html)

The page has three zones:

Zone	Contents
Header bar	App name 'AstraNotes', search bar, '+ New Note' button
Note list	Cards sorted newest-first. Each card shows: title, body preview, tags, timestamps, UUID, private lock icon
Create / Edit modal	Form with: Title (required), Body, Tags (comma-separated), Private checkbox. Inline validation error if title is empty

6. Note Card Data Fields

Each note card in the list must display the following fields, satisfying FR-06:

Field	Source	Requirement
Title	note.title	FR-01
Body preview	note.body[:200]	FR-01
Tags	note.tags (comma-separated)	FR-06
Created	note.created_at (UTC)	FR-06
Last modified	note.modified_at (UTC)	FR-06, FR-08
ID	note.id (UUID)	FR-06
Private lock icon	note.is_private == True	FR-04

7. Error Handling

All `ValidationError`, `NoteNotFoundError`, and `StoragePersistenceError` exceptions raised by `NoteService` are caught at the Flask route level and surfaced as user-facing flash messages — never as stack traces (SPR-02). Empty title on create shows an inline form error. Delete of a non-existent ID shows a flash error and reloads the list.

8. New Dependencies

Library	Version	Purpose	License
Flask	>=3.0	Web framework — routes, templates, request handling	BSD-3-Clause

Flask is the only new dependency. It will be added to `requirements.txt` with its version pin. All other libraries already present are unchanged (SPR-04).

9. How to Run

```
# from AstraNotes/src/
python3 -m astranotes.web.app

# then open in browser:
http://localhost:5000
```

The CLI remains fully functional and unchanged. Both interfaces share the same `~/.astranotes/data/` store.

